Copyrighted Materials
Copyright © 1994 Oxford University Press Retrieved from www.knovel.com

16

THREE-DIMENSIONAL FLOWS

16.1 General considerations

Homogeneous (or zero-dimensional), one-dimensional, two-dimensional, and axially symmetric flows are idealizations and all real gas flows are three-dimensional. In DSMC computations of the idealized flows with restricted dimensions, the other dimension or dimensions may be regarded as being so small that the statistical fluctuations correspond with those in the real gas. This is not possible in a three-dimensional flow and the number of real molecules that are represented by each simulated molecule (the parameter FNUM in the DSMC programs) must be regarded as a computational approximation. While this poses an ideological difficulty, it is the sheer magnitude of the computational task that is the most serious problem associated with DSMC computations of three-dimensional flows.

The various classes of grid that were discussed in §14.1 in the context of two-dimensional flows may be extended to three dimensions. The simplest possible grid consists of uniform rectangular parallelepipeds defined by three sets of equally spaced planes. The moving molecules are readily indexed to such a grid and, while three-dimensional quadric surfaces may be embedded in it, the cells can be 'fitted' only to flat surfaces that lie in one of the planes that define the cell boundaries. The demonstration program DSMC3 is an extension of the two-dimensional program DSMC2 and is an implementation of this simplest possible case. While some of the problems that can be studied through this program are of engineering interest, more sophisticated programs are required for most applications.

The two-dimensional 'general scheme' that was discussed as item (vi) of §14.1 has been extended to cover three-dimensional flows. The multiple four-sided regions that were the building blocks for that scheme become regions with six faces as illustrated in Fig. 16.1. The upper and lower faces are defined by a set of points that are specified by their x- y– and z–coordinates and are arranged in a rectangular array of rows and columns. There are an equal number of points in the rows and columns on each of these faces and the corresponding points are joined by straight lines. Each of these lines is divided into an equal number of points, and the corresponding points on each line are joined in order to define a regular three-dimensional array of six-faced cells. These cells may be referred to as 'deformed hexahedra', but they differ from a hexahedron in that the faces are, in general, not plane. This means that they cannot be used in tracing the progression of the molecules through the flow and they must be subdivided into tetrahedra.

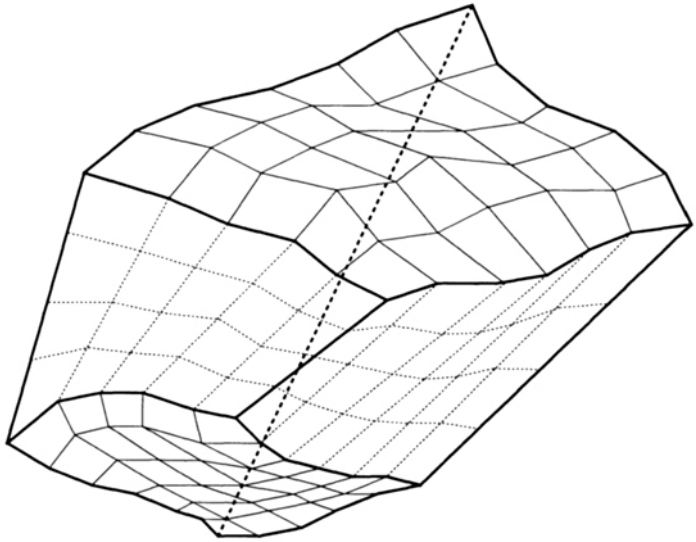


Fig. 16.1 An irregularly shaped six-faced region with an ordered array of cell columns, rows and slices.

Two of the faces of each region (the upper and lower in Fig.16.1) may have double curvature because there is no restriction on the locations of the array of points that define it. The points on the other faces must lie along a series of straight lines and the geometry of these faces is effectively limited to single curvature. Each face may be a solid surface, a plane of symmetry, an interface with a uniform or non-uniform flow, a vacuum, or an interface with other regions. Despite the restriction on the degree of curvature of four of the sides, the spatial domain of almost all problems can be divided into regions of this type. The molecules are traced through the flow by calculating the successive collisions of the molecules with the faces of the tetrahedra.

After the division into tetrahedra, the region boundaries become a set of triangular plane elements. Each of the deformed hexahedra may be divided into one 'central' and four 'corner' tetrahedra as shown in Fig. 16.2. Alternatively, each hexahedron may be divided across one of the 'diagonal planes' into two solid elements with three four-sided and two triangular faces. Each of these elements may then be divided into three tetrahedra, as shown in Fig. 16.3 for the right-hand-side element. The 'six-tetrahedron scheme' is to be preferred over the 'five-tetrahedron scheme' which can break down for the very thin hexahedra that are often required in regions with high flow gradients. The breakdown occurs when the tetrahedron edges that join the opposite corners of one face intersect with the opposite face. In addition, the tetrahedra serve as sub-cells and there is less average separation of the collision pairs in the alternative scheme.

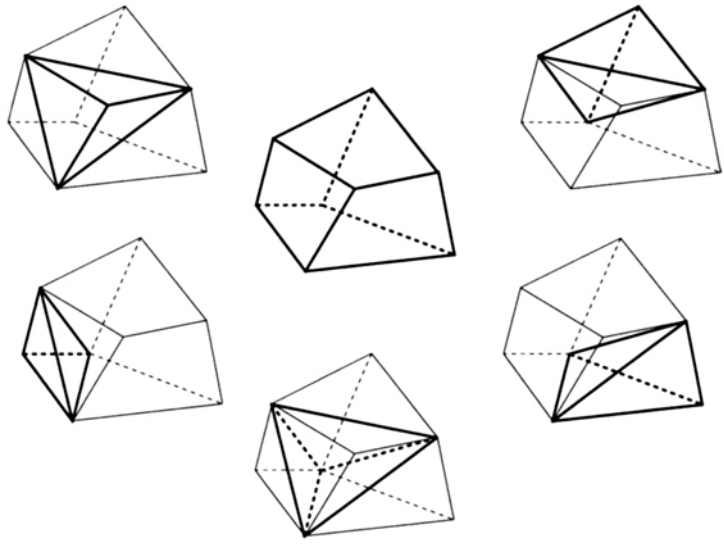


Fig. 16.2 The subdivision of a 'deformed hexahedral' cell into five tetrahedral sub-cells.

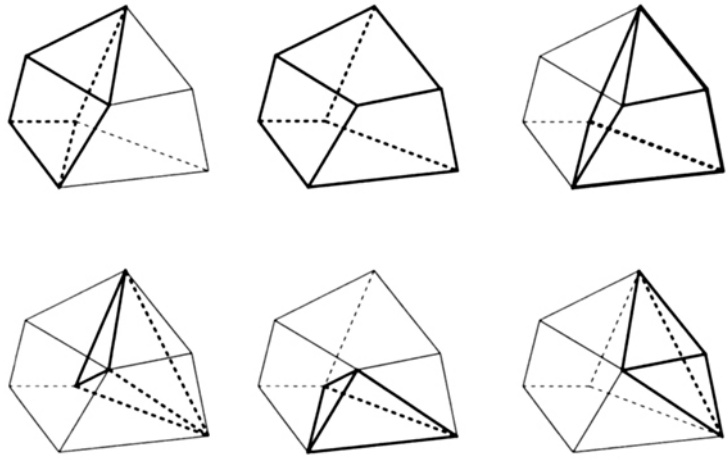


Fig. 16.3 The subdivision of a 'deformed hexahedral' cell into six tetrahedral sub-cells.

As noted earlier, the indexing of each molecule to the cells requires the calculation of the intersection point of its trajectory with each tetrahedral face through which it passes. This is not only expensive in terms of computation time, but round-off error is a serious problem. It can be seen from Fig. 16.3 that up to 24 tetrahedra can meet at a single node of the grid. When a molecule passes extremely close to a node, the round off error may place it into the wrong tetrahedron and it then becomes 'lost'. It has been found that, if lost molecules are simply discarded from the calculation, the resulting error can be unacceptable unless double precision arithmetic is employed. It may be necessary to introduce 'recovery' procedures to deal with these lost molecules and to also cope with the gaps that can develop in the flowfield if there is a change in grid spacing at an interface between regions.

This scheme has been employed in a number of studies (e.g. Celenligil and Moss 1992), but the labour that is necessary for the generation of the mass of data that is required for the specification of a complex application can pose a problem that is even more serious than the computation time. Some form of automatic grid generation is necessary in practice and the protocols that must be followed in the specification of multiple regions leads to difficulties with the logic. The objective is to develop a code that requires only the specification of the boundaries, either as sets of coordinates or analytical expressions, and the flow conditions. The program *should itself generate the grid and, ideally, it should adapt the grid to its* optimal form as the flow develops.

One possible solution is to replace the structured tetrahedral grid by an unstructured tetrahedral grid of the type that has been widely used in applications of the finite element method. Each triangular face within the grid either lies on a boundary or is an interface between two adjacent tetrahedra. The molecules are again followed through their intersections with these triangular faces. The challenge is to develop procedures that efficiently generate a near-optimal grid and to minimize the amount of information that must be stored for each tetrahedral element.

Another option is to extend to three-dimensions the scheme that was described under heading (viii) in §14.1. That scheme employed cells that were aggregated from small sub-cells, with the surface being described by small straight line segments in the appropriate sub-cells. However, it is difficult to extend this to three dimensions because of the complexity of the possible interaction of surfaces with three-dimensional sub-cells. An alternative is to subdivide the sub-cells through which the surfaces pass into much smaller rectangular elements and to define the surface by marking some of these elements as 'surface elements'. While this causes the surface to have a stepped structure, the correct direction cosines of the surface normal may be stored for each element, and the approximation that is introduced through this process then becomes one of resolution rather than of shape. This scheme should lead to satisfactory results as long as the surface elements are smaller than the local mean free path in the flow.

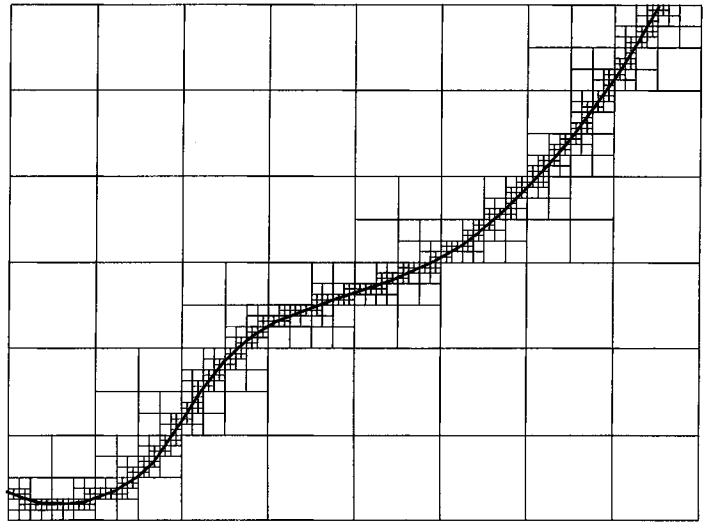


Fig. 16.4 The division of sub-cells to a depth of four for the definition of surface elements.

This scheme has been implemented (Bird 1990) for the Shuttle orbiter geometry. The sub-cell array was 60×48×42 and the sub-cells containing surface elements were divided 8×8×8. The linear dimension of a surface element was approximately 10 cm and, as expected, the error became significant at about 120 km altitude where the mean free path falls to 10 cm. This surface definition required only 2 Mb of memory but, to reduce the linear dimension of a surface element by a factor of ten to maintain accuracy to 15 km lower altitude, requires the number of surface elements to increase by a factor of 1000. The solution is to divide a cell into eight equal parts, then divide only those parts that include the surface into eight further parts, and so on. This procedure is illustrated in two dimensions in Fig 16.4. This scheme has been implemented to a depth of eight divisions such that the linear dimension of the element is reduced by a factor of 256. It was verified that the storage requirements for this scheme increase as the inverse square, rather than the inverse cube, of the element size. Only one integer is required for each of the elements at any depth of subdivision and, at the surface element level, the direction cosines and surface temperature are packed into this integer. Sub-cells and elements that do not include surface elements are also divided where necessary to produce smaller cells. The disadvantage of this scheme is that it is very complex and, while it permits much smaller elements, they retain a finite size and there are unavoidable side effects associated with this. For example, the edge of a surface cannot be thinner than the linear dimension of a surface element. The optimal grid scheme for engineering studies of complex three-dimensional flows of rarefied gases has yet to be determined.

16.2 Supersonic corner flow

The data in the disk listing of program DSMC3 is for the flow in the 'corner' between two flat plates that are perpendicular to one another and parallel to the stream. It is a basic three-dimensional flow that is related to the two-dimensional leading edge or flat plate problem that was the main test case for program DSMC2 in §14.3.

As noted above, program DSMC3 is a straightforward extension of program DSMC2 to three dimensions. The cell 'columns' in the x-direction and 'rows' in the y-direction are supplemented by cell 'slices' in the z-direction. The flowfield is a rectangular parallelepiped and each of its six faces may be either a plane of symmetry, an interface with a uniform stream, or a vacuum. The program does not have the option of a geometric progression for the cell size, and the cells must have a uniform width, height, and depth in the x-, y-, and z-directions, respectively. The surfaces must be plane and lie along an interface between cell rows, columns, or slices. They must be rectangular in shape and must terminate at cell boundaries. The surfaces may lie along a flowfield boundary as long as the boundary is otherwise a plane of symmetry. A uniform jet may be present in addition to the external stream, but its dimensions are no longer tied to the cell boundaries and it must be circular in cross-section.

The two-dimensional calculation in §14.3 employed a 100×60 cell grid with four sub-cells per cell. The three-dimensional calculation is of a similar computational magnitude if the cell grid is 30×18×18 with no sub-cells. The stream is argon at a temperature of 300 K and a number density of 10^{20} m^{-3} at a Mach number of 6, and is parallel to the x-axis. The width, height, and depth of the cells is 0.01 m and the 30 cells are along the x-axis. The leading edge of the plates is five cells from the upstream face of the flowfield at $x = 0$ and the surfaces cover the remainder of the y=0 and $z = 0$ planes. Diffuse reflection is assumed and the surfaces are at a uniform temperature of 1000 K. The freestream mean free path is 0.0129 m and the cell size is greater than the recommended value for DSMC computations.

Program DSMC2 was used to make calculations for the corresponding two-dimensional flow to which the three-dimensional flow can be expected to tend at large distances from the corner. Because of the marginal cell size in the three-dimensional calculation, the two-dimensional calculation was first made for a 30×18 grid with no sub-cells that corresponds to the three-dimensional grid. It was then repeated for a 60×36 grid with two sub-cells per cell in each direction. The latter grid easily meets the requirements of the DSMC method.

The distribution of the pressure coefficient over the plates is shown in Fig. 16.5. The flow is symmetrical about a plane that includes the x-axis and bisects the angle between the plates. This means that the contours on each plate should ideally be identical and they are, in fact, very similar. The results from the corresponding two-dimensional calculations for the distribution of the pressure coefficient are shown in Fig. 16.6.

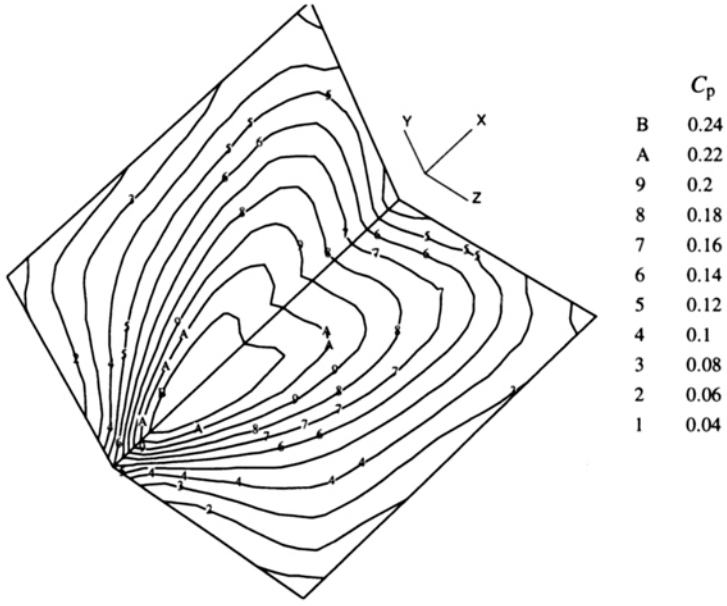


Fig. 16.5 Contours of pressure coefficient over the surfaces.

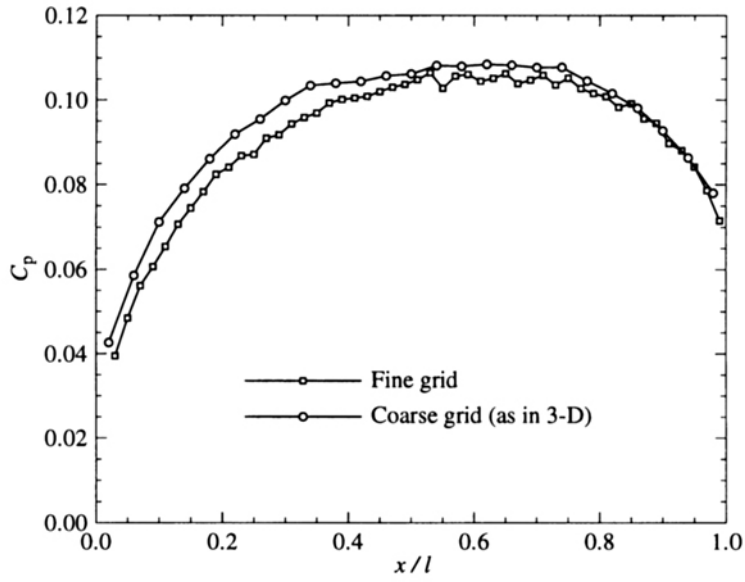


Fig. 16.6 The distribution of pressure coefficient over the corresponding two-dimensional flat plate.

The length l of the plate is 23.26 freestream mean free paths so that the overall Knudsen number is 0.043. The two-dimensional pressure distribution has a maximum at about 15 freestream mean free paths from the leading edge. A comparison with Fig. 14.12, which is for a diatomic gas at $(Ma)=4$ and $(Kn)=0.0143$, shows that the pressure maximum is now further from the leading edge. As expected from the study of cell size in §14.4, the coarse grid that is used in the three-dimensional calculations causes the pressure over the forward half of the plate to be up to 8% too high. 'This systematic error should be kept in mind when interpreting the three-dimensional results. The boundary condition at the outer planes that are defined by $y=0.18$ and $z=0.18$ is an undisturbed freestream rather than a plane of symmetry, and this would be expected to lead to some reduction in the pressure near these boundaries. The pressure distribution along these outer boundaries is about 20% less than the two-dimensional pressures.

The pressure coefficient along the junction between the plates has a maximum at about $x/l=0.3$. This maximum is also the maximum pressure on the plates and its value is just twice the maximum pressure in the two-dimensional case. The pressure at the trailing edge of the junction is less than 50% greater than the trailing edge pressure in the two-dimensional case. The downstream boundary is specified as a stream boundary and, at this Mach number, there is effectively no difference between this and a vacuum boundary. It is possible that the effects of this boundary condition extend further upstream in the corner than in the outer part of the flow or in the two-dimensional case.

The corresponding three-dimensional and two-dimensional results for the net heat transfer coefficient are shown in Figs 16.7 and 16.8. The effect of cell size is similar to that for the pressure coefficient. The heat transfer coefficient at the outer edge is only about 10% less than in the two-dimensional case. It increases towards the junction between the plates, but there is a reduction very close to the junction and the maximum heat transfer occurs in two lobes that are some mean free paths from the junction. The maximum heat transfer is less than 50% higher than the maximum two-dimensional heat transfer, so that the proportional increase is less than that for the pressure. The reduction close to the junction becomes more pronounced towards the trailing edge and the heat transfer at the trailing edge of the junction is only one third of the minimum heat transfer to the two-dimensional plate.

The results for the skin friction coefficient based on the component of shear stress in the x-direction are shown in Figs 16.9 and 16.10. The effect of cell size is similar to that in the other cases, but the skin friction at the outer edge shows almost no reduction on the two-dimensional distribution. The general form of the contours is remarkably similar to that for the net heat transfer coefficient. The values of the skin friction coefficient are of the order of twice the net heat transfer coefficient. This indicates that the 'Reynolds analogy' between heat transfer and skin friction may sometimes be a useful approximation for rarefied gas flows.

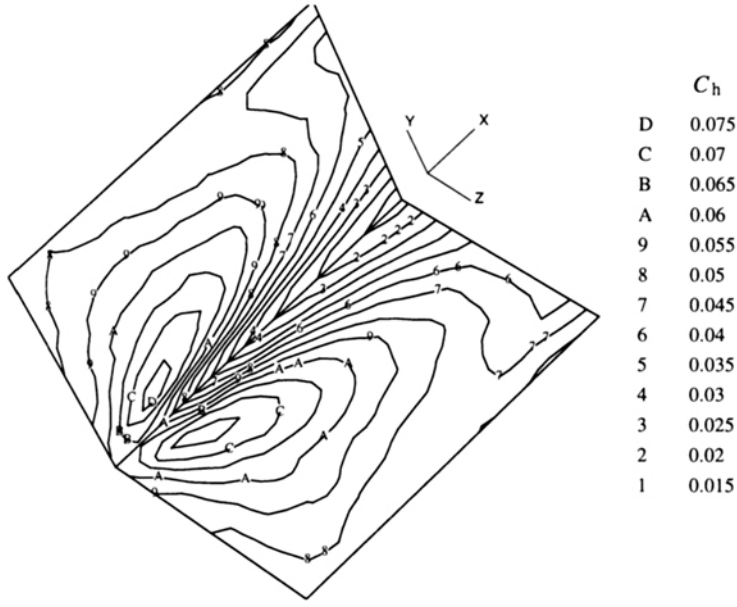


Fig. 16.7 Contours of constant value of the heat transfer coefficient in the three-dimensional case.

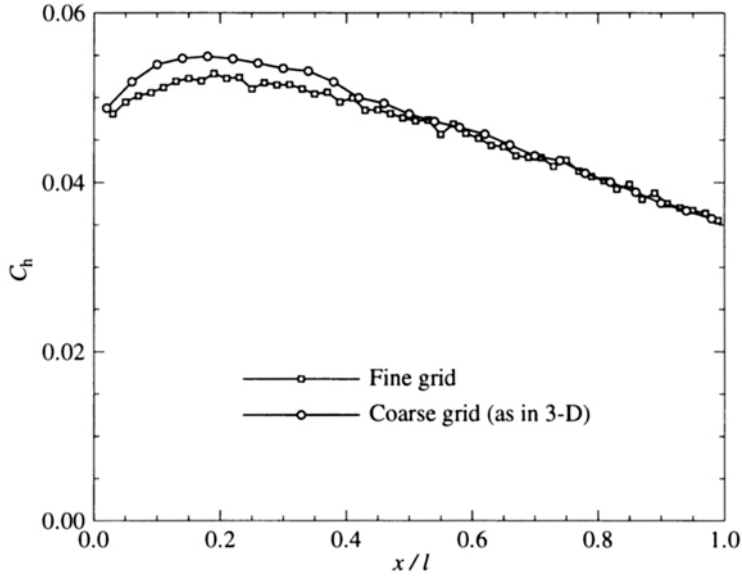


Fig. 16.8 The heat transfer distribution in the two-dimensional case.

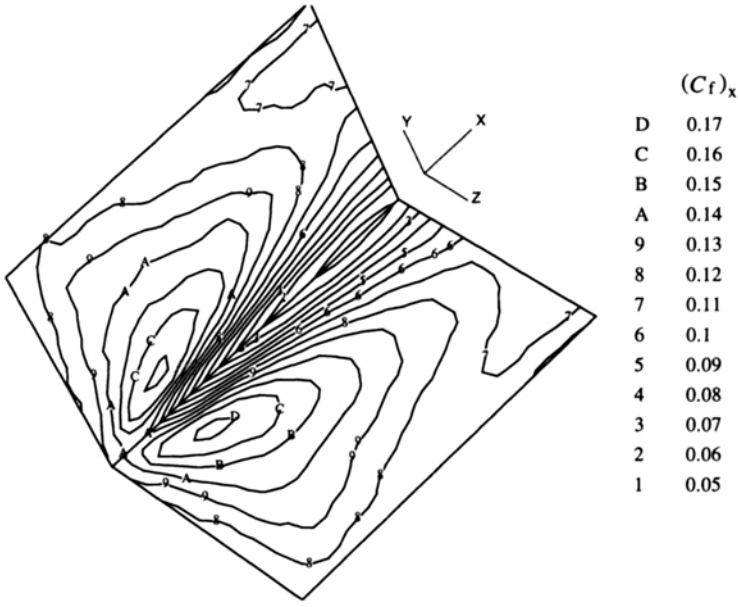


Fig. 16.9 The streamwise skin friction coefficient in the corner flow.

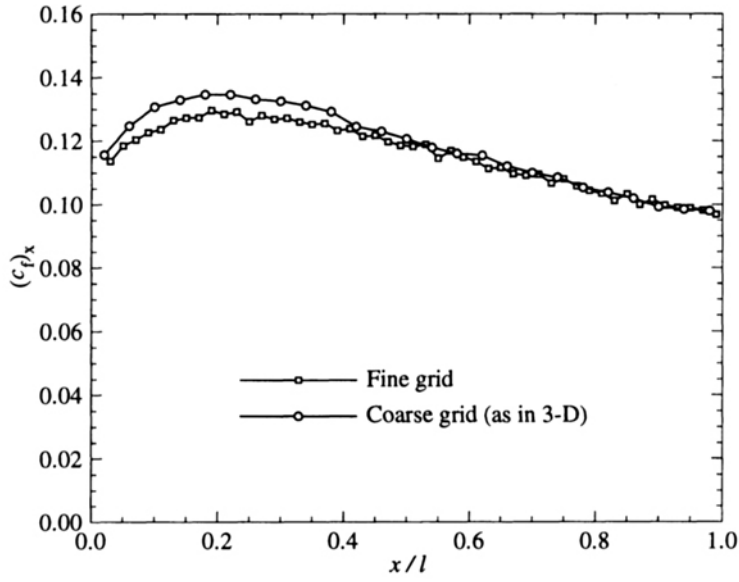


Fig. 16.10 The distribution of the streamwise skin friction coefficient in the two-dimensional case.

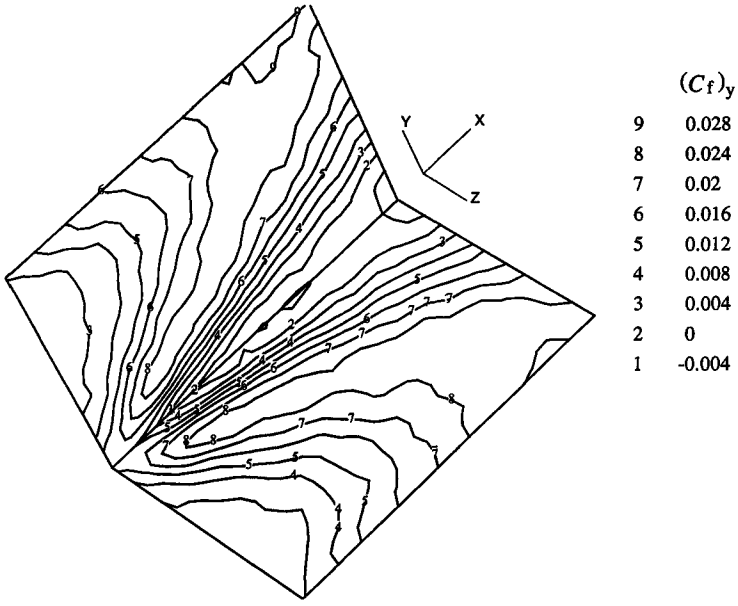


Fig. 16.11 The normal skin friction coefficient in the corner flow.

The skin friction coefficient in the direction normal to the streamwise direction is denoted by $(c_f)_y$ and the contours of constant value of this coefficient are shown in Fig. 16.11. As would be expected, this coefficient increases towards the trailing edge and toward the outer part of the plate. The flow near the junction of the plates appears to be directed toward the junction over most of the span, but the negative skin friction coefficients are very small in magnitude. The magnitude of the normal coefficient is generally small in comparison with that of the streamwise skin friction coefficient.

Finally, Figs 16.12 to 16.14 show, respectively, contours of constant value of temperature, density, and Mach number in a number of planes normal to the streamwise direction. The density contours are qualitatively different from the temperature and Mach number contours, and the most interesting feature of these contours is the isolated region of high density that develops on the plane of symmetry. The contours in the plane nearest the leading edge approach the expected two-dimensional configuration in the outer region of the flow, but those near the trailing edge show that the computational grid would have to be very much larger if the transition to a two-dimensional flow was to be adequately covered. The gas in the vicinity of the corner has a high temperature and density, but a very small velocity. The slip velocity increases at the outer edge of the plate and this is associated with the 'stream' boundary condition. Low subsonic velocities occur only in a small region near the corner.

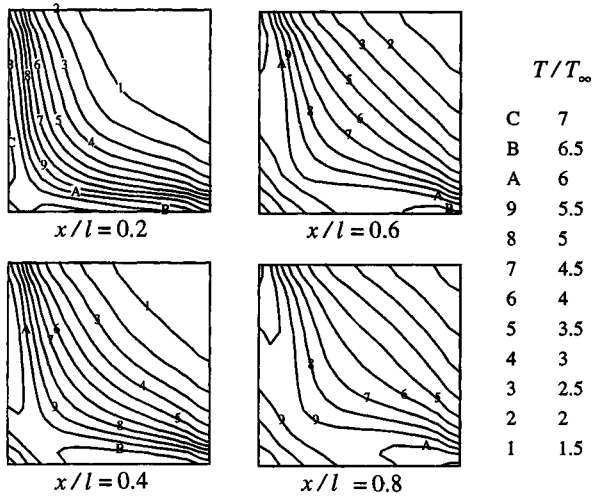


Fig. 16.12 Temperature contours in planes normal to the plates.

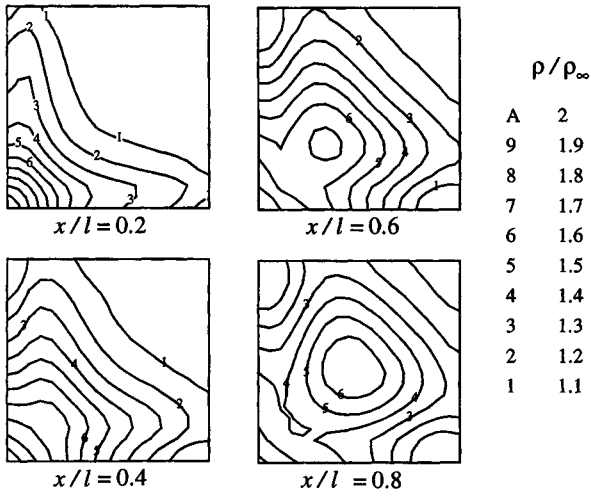


Fig. 16.13 Density contours in planes normal to the plates.

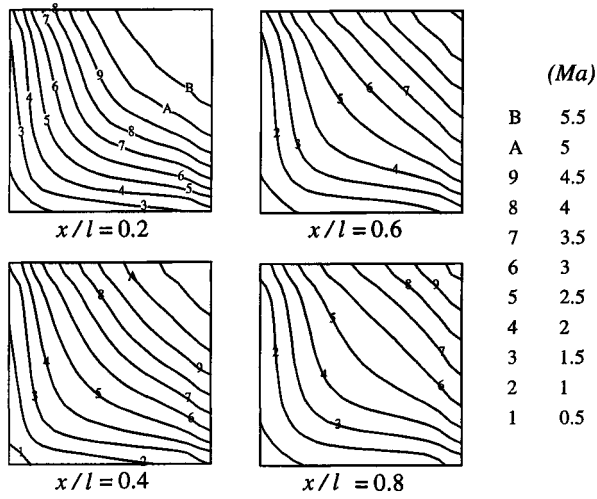


Fig. 16.14 Mach number contours in planes normal to the plates.

16.3 Finite span flat plate at incidence

The flow of argon at a Mach number of 5 about a two-dimensional flat plate was studied in §14.8. The main test case was for an incidence of 30° and an overall Knudsen number of 0.043. This Knudsen number is too low for a three-dimensional calculation of the magnitude to which these test cases are limited. However, the 30×18×18 grid that was used in the previous application of program DSMC3 allows calculations for a three-dimensional plate with an aspect ratio of two at an overall Knudsen number of 0.108. The stream direction is normal to the z-axis and the z =0 plane is a plane of symmetry. The other boundaries are interfaces with the uniform stream. The double-sided surface that forms the flat plate lies along the lower boundary of the ninth cell row. It extends from the start of cell column 10 to the end of column 21 in the x-direction and from the plane of symmetry to the end of slice 12 in the z-direction. The half of the plate that is in the computational region therefore has a square planform. As in the two-dimensional case, the temperature of the upstream or lower side of the plate is equal to the stream stagnation temperature, and the downstream or upper side is at the stream static temperature. The surface reflection is diffuse with complete thermal accommodation. The high temperature of the lower surface minimizes the density increase, but the cell size is again larger than is recommended for DSMC computations.

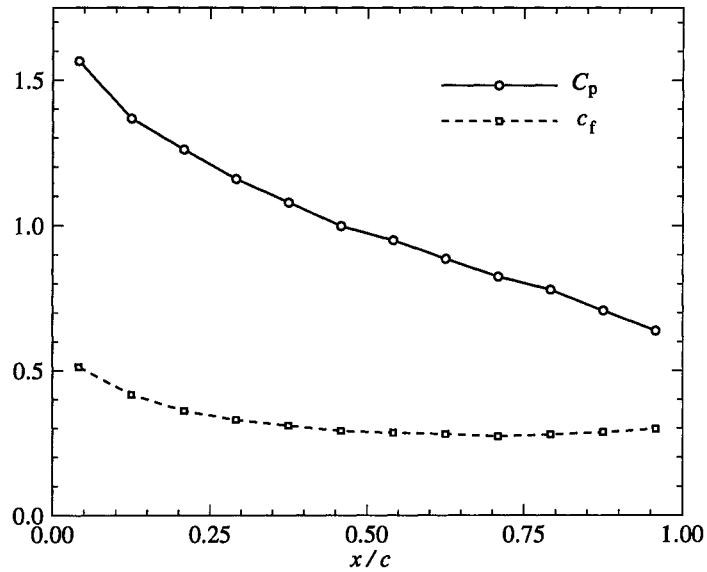


Fig. 16.15 Contours of the pressure coefficient over the lower surface.

The pressure distribution over the lower surface is shown in Fig. 16.15. The corresponding results for the streamwise and spanwise skin friction coefficients are shown in Figs 16.16 and 16.17, respectively. The distributions of pressure and skin friction that are obtained from program DSMC2 for the corresponding two-dimensional flow are shown in Fig 16.18. The two-dimensional calculation employs a grid with the same resolution, so that the errors due to the coarser than recommended grid should be similar. There is good agreement between the three-dimensional results at the plane of symmetry and the two-dimensional results. The increase in the skin friction coefficient near the trailing edge is due to the acceleration of the flow in this region, but it is less pronounced than in Fig. 14.44 for a lower Knudsen number.

The effects due to the finite span show primarily as gradients in the spanwise direction, and these increase with proximity to the tip. However, at the trailing edge, there appears to be a small reduction in pressure and the development of a spanwise shear stress component that persists almost to the plane of symmetry. The major effect on the pressure distribution is a reduction in the pressure near the leading edge and towards the tip. The relative change in the streamwise shear stress is slightly larger than that for the pressure. The spanwise skin friction coefficient becomes significant near the tip, but it is small in comparison with the streamwise coefficient.

Finally, Fig. 16.19 shows some contours of constant Mach number in planes over the plate that are normal to the stream direction.

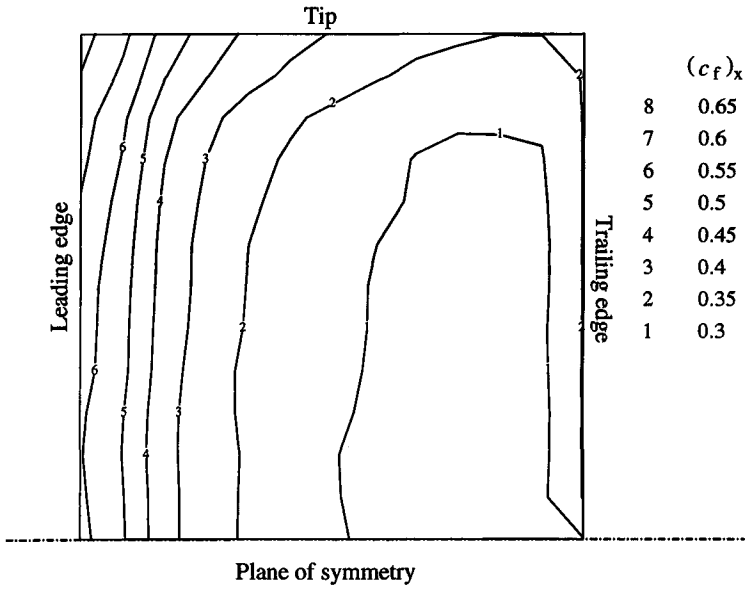


Fig. 16.16 The streamwise skin friction coefficient on the lower surface.

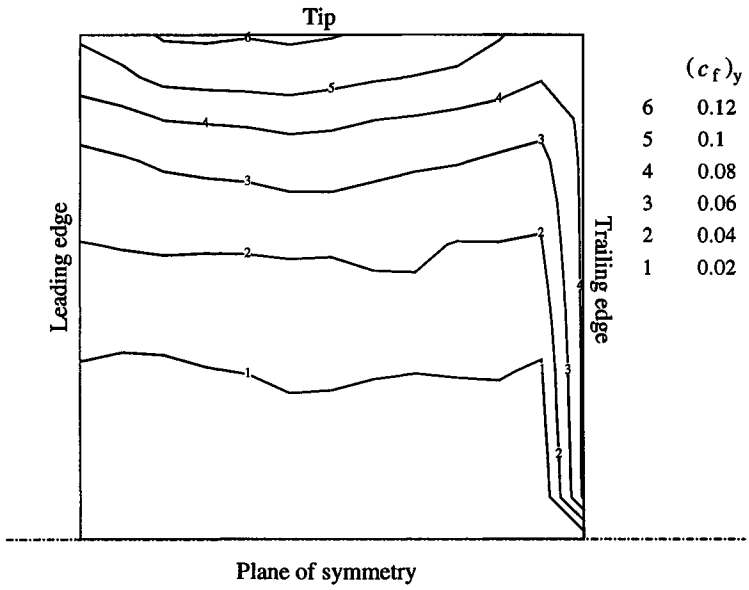


Fig. 16.17 The spanwise skin friction coefficient on the lower surface.

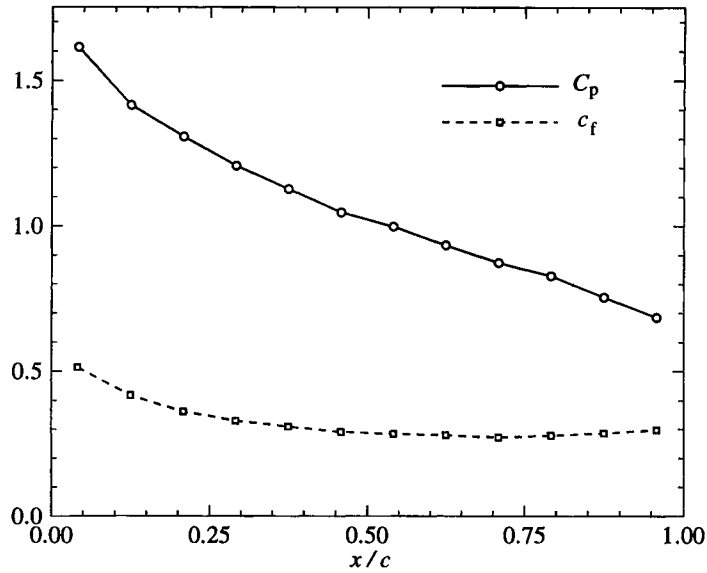


Fig. 16.18 The pressure and skin friction distribution along the lower surface of the corresponding two-dimensional flow.

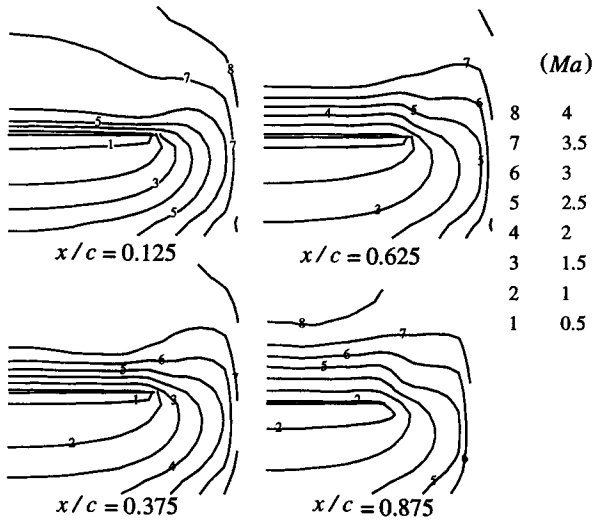


Fig.16.19 Mach number contours in planes normal to the stream.

16.4 Corner flow with normal jet plume

Click for referenced Color Plate

The final demonstration flow takes advantage of the jet option in program DSMC3 to add a jet plume to the supersonic corner flow of §16.2. The constraints on the data force the exit plane of the jet to lie in the $y = 0$ plane and the exit flow must be in the positive y-direction. The radius of the jet is 0.03 m and its centre is at $x = 0.15$, $z = 0.09$. This location is at 40% of the length of the surface and midway between the junction between the surfaces and the outer boundary. The exit plane density is three times the stream density, while its composition, temperature, and speed are equal to the corresponding stream values. No other changes were made to the data that led to the results in §16.2 for the supersonic corner flow.

The addition of the plume causes the corner flow to lose its symmetry about the plane that bisects the angle between the horizontal and vertical surfaces. The contours of constant value of the pressure coefficient, the resultant (rather than the components of the) skin friction coefficient, and the heat transfer coefficient are shown in Figs 16.20 to 16.22. The extent to which the jet modifies the flow may be seen from a comparison of these figures with Figs 16.5 to 16.11. The most striking feature is the region of high pressure, shear stress, and heat transfer that develops on the vertical surface. The distribution of heat transfer is again similar to that for the shear stress.

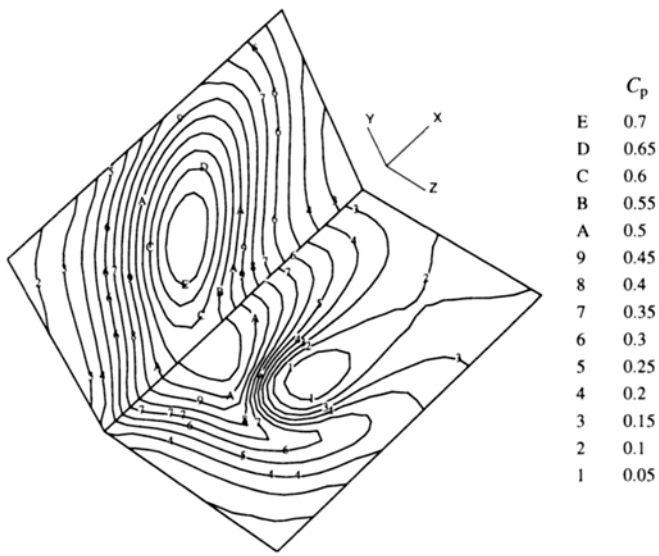


Fig. 16.20 The corner flow pressure distribution in the presence of a jet with its efflux plane in the xz-plane.

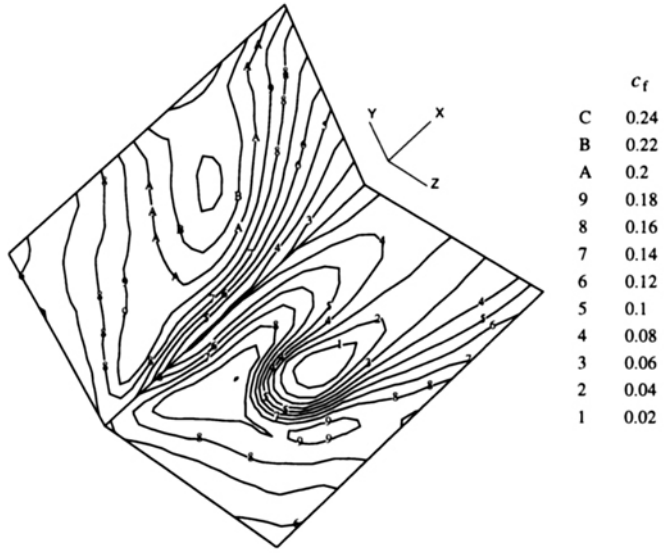


Fig. 16.21 The distribution of the skin friction coefficient.

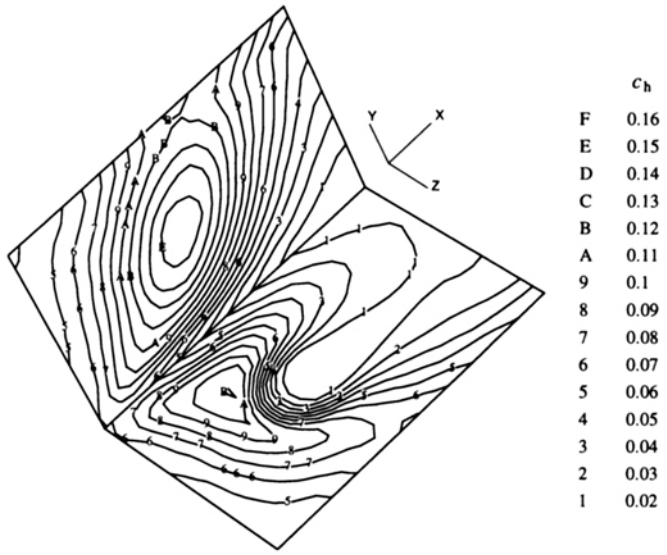


Fig. 16.22 The distribution of the heat transfer coefficient.

16.5 Concluding remarks

The applications of the demonstration programs have been limited to cases that require no more than 24 hours of computation time on a contemporary 'top of the line' personal computer. This has been taken to be an 'IBM compatible' machine equipped with an Intel i486 running at 66 Mhz. In addition, the problems have been restricted to a memory requirement of 6 Mb. By the time that this book is published, the speed and memory capacity of personal computers will have doubled. This doubling of capability has, over the past fifteen years, occurred at intervals of about two years and this rate of progress can be expected to continue.

The computational limits were not a significant factor in the choice of problems for homogeneous and one-dimensional flow problems. These flows can be calculated with a sufficiently high sample size that the statistical scatter is reduced to less than one per cent. Moreover, test calculations can be made for conditions that fall within the range of validity of the continuum equations.

Two-dimensional calculations at 'continuum Knudsen numbers' require computer resources that exceed the limits that have been applied here to the demonstration cases.

Three-dimensional calculations are much more difficult and it is not yet possible to make three-dimensional calculations for non-trivial applications in the continuum regime. The degree of difficulty that is now associated with these calculations is, however, comparable to that which applied to two-dimensional flows twenty years ago.

'Engineering' applications are still largely restricted to cases with local Knudsen numbers that are sufficiently high for the problem to fall within the bounds of 'rarefied gas dynamics'. However, should computer capacity continue to increase by an order of magnitude every five years and DSMC affordability increase by a further factor of one million over the next thirty years, the DSMC method can be expected to make significant inroads into the realm of continuum CFD.

References

BIRD, G.A. (1990). Application of the direct simulation Monte Carlo method to the full Shuttle geometry, Am. Inst. Aero. Astro., Paper AIAA-90-1692.

CELENLIGIL, M.C. and MOSS, J.N. (1992). Hypersonic rarefied flow about a delta wing – direct simulation and comparison with experiment, *AIAA Journal*, **30**, 2017–2023.